Lines, angles, and triangles — Part 1 answer key

#	ID	Answer	Difficulty
1.1	4d53ed15-713b-4aa2-8943-17060404b617	B	Easy
1.2	32a77034-9666-46e7-b78c-a2b1427d5d9b	D	Easy
1.3	823ebf16-5273-4421-aeea-7ddb20cca4fd	47	Easy
1.4	2e29358e-5d5c-485d-bea8-cb085eba5056	D	Easy
1.5	cbf53556-86ef-493b-a586-486445620deb	B	Easy
1.6	8d0aca2e-6014-42d2-adde-c7e271bac2be	162	Easy
1.7	af20e9d4-dd92-4ae0-9ead-3798a3e06b80	A	Easy
1.8	2b0e8838-4d1b-4eaf-9a92-439aa05ee874	B	Easy
1.9	c6b681f6-504a-4e3a-8c73-a1cc8632d86b	D	Easy
1.10	d14a2fdd-4d09-4911-b363-67cb2fe299e3	B	Easy
1.11	029285-DC	C	Easy
1.12	029330-DC	A	Easy
1.13	b693eec7-7a41-4127-adf6-bb6bf263019b	A	Easy
1.14	08250-DC	C	Easy
1.15	029526-DC	?	Easy
1.16	08194-DC	A	Easy
1.17	7bf2c7eb-d2eb-49db-a4cf-7cd376347a2c	D	Easy
1.18	027075-DC	D	Easy
1.19	2d07d31c-7627-4c6e-813d-60c0e28346b7	A	Easy
1.20	7d655176-161a-479c-b588-adb3cf6c69c2	B	Easy

1.21	14890-DC	B	Easy
1.22	50ce06e1-35f6-477e-ac29-35d74d5ab3b6	175	Easy
1.23	96a4ec26-6ad9-4b3b-9ae1-693feee5691e	D	Easy
1.24	af58075c-8456-4696-8179-6c1a86044478	C	Easy
1.25	019566-DC	B	Easy
1.26	a79dbc7d-8d01-40c4-8900-15940eeff7d7	A	Easy
1.27	095e8ea4-b798-4679-8448-1de5fc94090d	D	Easy
1.28	5ca08c31-e18f-4cb7-837a-0814ebea3a8e	A	Easy
1.29	026874-DC	B	Easy
1.30	9df9e1c7-bfff-4d4e-865f-5cbcb4b24e4c	D	Easy
1.31	baefd1d0-91cf-4b82-9c41-2145510636af	A	Easy
1.32	b09e0918-6dfc-4b50-b503-bbdd89fe3480	70	Easy
1.33	dbae1ccb-c727-453d-aad3-d9808097dcfd	B	Easy
1.34	3f1bac86-dd93-4b6b-960c-c29978b05446	B	Easy
1.35	bafc5567-f967-4844-ba8a-9cd3d5bc11de	B	Easy
1.36	022230-DC	C	Easy
1.37	c6d50efa-46fb-4406-9c84-8c8d2ab9e927	C	Easy
1.38	60ddced0-ee94-414c-9143-3704d3c18d2e	A	Easy
1.39	b3738b72-9c5a-45b5-986b-00bb9643ab90	C	Easy
1.40	6145108f-5273-4b08-afde-8e75072e2eef	D	Easy
1.41	027209-DC	?	Medium
1.42	551e10b5-2f0d-4002-a399-823a818766b9	C	Medium
1.43	6a820c4b-cc06-4343-a5bd-fe398a0d2ec6	C	Medium
1.44	b1f0e2ca-1e0c-430d-b61d-6cf8594501a5	A	Medium
1.45	64da6245-4314-4da9-817e-46dcdd9dc29d	B	Medium
1.46	029403-DC	B	Medium

1.47	e8922caa-569e-45e5-9744-1704932fd187	21	Medium
1.48	275a2867-6329-41e4-8bab-072a1e2475e7	A	Medium
1.49	2fc71dbf-343a-44f3-80ee-5b7f22a564da	B	Medium
1.50	9fe57ebf-964f-4add-9e16-f54cf44c7bf4	A	Medium
1.51	e2fa4379-f216-4286-8c00-6098b985ceea	D	Medium

Rationales

▼ 1.1 — B

Choice B is correct. It's given that triangle *ABC* is congruent to triangle *DEF*. Corresponding angles of congruent triangles are congruent and, therefore, have equal measure. It's given that angle *A* corresponds to angle *D*, and that the measure of angle *A* is $18°$. It's also given that the measures of angles *B* and *E* are $90°$. Since these angles have equal measure, they are corresponding angles. It follows that angle *C* corresponds to angle *F*. Let $x°$ represent the measure of angle *C*. Since the sum of the measures of the interior angles of a triangle is $180°$, it follows that $18° + 90° + x° = 180°$, or $108° + x° = 180°$. Subtracting $108°$ from both sides of this equation yields $x° = 72°$. Therefore, the measure of angle *C* is $72°$. Since angle *C* corresponds to angle *F*, it follows that the measure of angle *F* is also $72°$.

Choice A is incorrect. This is the measure of angle *D*, not the measure of angle *F*.

Choice C is incorrect. This is the measure of angle *E*, not the measure of angle *F*.

Choice D is incorrect. This is the sum of the measures of angles *E* and *F*, not the measure of angle *F*.

▼ 1.2 — D

Choice D is correct. It's given that line *m* is parallel to line *n*, and line *t* intersects both lines. It follows that line *t* is a transversal. When two lines are parallel and intersected by a transversal, exterior angles on the same side of the transversal are supplementary. Thus, $x + 33 = 180$. Subtracting 33 from both sides of this equation yields $x = 147$. Therefore, the value of *x* is 147.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.3 — 47

The correct answer is 47. Based on the figure, the angle with measure $x°$ and the angle with measure $133°$ together form a straight line. Therefore, these two angles are supplementary, so the sum of their measures is $180°$. It follows that $x + 133 = 180$. Subtracting 133 from both sides of this equation yields $x = 47$.

▼ 1.4 — D

Choice D is correct. In the figure shown, the angle marked $x°$ and the angle marked $142°$ form a linear pair of angles. If two angles form a linear pair of angles, the sum of the measures of the angles is $180°$. Therefore, the value of $x + 142$ is 180.

Choice A is incorrect. This is 90 less than 142, not the sum of x and 142.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the measure, in degrees, of one of the angles shown.

▼ 1.5 — B

Choice B is correct. In the figure shown, the angles with measures $x°$ and $72°$ are vertical angles. Vertical angles have equal measure, so the two marked angles have equal measure. Therefore, the value of x is 72.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.6 — 162

The correct answer is 162. It's given that line r is parallel to line s. Since line n intersects both lines r and s, it's a transversal. The angles in the figure marked as $162°$ and $x°$ are on the same side of the transversal, where one is an interior angle with line s as a side, and the other is an exterior angle with line r as a side. Thus, the marked angles are corresponding angles. When two parallel lines are intersected by a transversal,

corresponding angles are congruent and, therefore, have equal measure. It follows that the value of x is 162.

▼ 1.7 — A

Choice A is correct. In the figure shown, lines j and k are parallel if and only if a pair of corresponding angles are congruent. It's given that one angle has a measure of $40°$ and that the corresponding angle has a measure of $w°$. Therefore, $w = 40$ is sufficient to prove that lines j and k are parallel.

Choice B is incorrect. The angle measuring $x°$ and the angle measuring $40°$ are alternate interior angles. Thus, if lines j and k are parallel, x is equal to 40, not 140.

Choice C is incorrect. The angle measuring $y°$ and the angle measuring $40°$ are vertical angles. Thus, $y = 40$, whether lines j and k are parallel or not.

Choice D is incorrect. The angle measuring $z°$ is supplementary to the angle measuring $40°$. Thus, $z = 180 - 40$, or $z = 140$, whether lines j and k are parallel or not.

▼ 1.8 — B

Choice B is correct. The sum of the measures of the interior angles of a triangle is 180 degrees. Since the triangle is a right triangle, it has one angle that measures 90 degrees. Therefore, the sum of the measures, in degrees, of the remaining two angles is $180 - 90$, or 90. It's given that the measure of one of the acute angles in the triangle is 51 degrees. Therefore, the measure, in degrees, of the other acute angle is $90 - 51$, or 39.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the measure, in degrees, of the acute angle whose measure is given.

▼ 1.9 — D

Choice D is correct. In the figure shown, the angles with measures $w°$ and $z°$ are vertical angles. Since vertical angles are congruent, $w = z$. Therefore, if $w = 136$, the value of z is 136.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the measure, in degrees, of an angle that's supplementary, not congruent, to the angle with measure $w°$.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.10 — B

Choice B is correct. The triangle shown is a right triangle, where the interior angle shown with a right angle symbol has a measure of $90°$. It's shown that the other two interior angles measure $13°$ and $a°$. The sum of the measures of the interior angles of a triangle is $180°$; therefore, $90 + 13 + a = 180$. Combining like terms on the left-hand side of this equation yields $103 + a = 180$. Subtracting 103 from both sides of this equation yields $a = 77$.

Choice A is incorrect. This is the measure, in degrees, of the other acute interior angle of the right triangle, not the value of a.

Choice C is incorrect. This is the measure, in degrees, of the right angle of the right triangle, not the value of a.

Choice D is incorrect. This is the sum of the measures, in degrees, of the other two interior angles of the right triangle, not the value of a.

▼ 1.11 — C

Choice C is correct. The angle that is shown as lying between the y° angle and the z° angle is a vertical angle with the x° angle. Since vertical angles are congruent and $x = 65$, the angle between the y° angle and the z° angle measures 65°. Since the 65° angle, the y° angle, and the z° angle are adjacent and form a straight angle, it follows that the sum of the measures of these three angles is 180°, which is represented by the equation $65° + y° + z° = 180°$. It's given that y = 75. Substituting 75 for y yields $65° + 75° + z° = 180°$, which can be rewritten as $140° + z° = 180°$. Subtracting 140° from both sides of this equation yields $z° = 40°$. Therefore, $z = 40$.

Choice A is incorrect and may result from finding the value of $x + y$ rather than z. Choices B and D are incorrect and may result from conceptual or computational errors.

▼ 1.12 — A

Choice A is correct. The sum of the measures of the three interior angles of a triangle is 180°. Therefore, $31 + 2b + a = 180$. Since it's given that $a = 45$, it follows that

$31+2b+45=180$, or $2b=104$. Dividing both sides of this equation by 2 yields $b=52$.

Choice B is incorrect and may result from a calculation error. Choice C is incorrect. This is the value of $a+31$. Choice D is incorrect. This is the value of $2b$.

▼ 1.13 — A

Choice A is correct. In an equilateral triangle, all three sides have the same length. It's given that in triangle *ABC*, *AB* = 4,680 mm and *BC* = 4,680 mm. Therefore, if *AC* = 4,680 mm, then all three sides of triangle *ABC* have the same length, so triangle *ABC* is equilateral. Therefore, *AC* = 4,680 mm is sufficient to prove that triangle *ABC* is equilateral.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.14 — C

Choice C is correct. Since lines ℓ and k are parallel, corresponding angles formed by the intersection of line j with lines ℓ and k are congruent. Therefore, the angle with measure a° must be the supplement of the angle with measure 64°. The sum of two supplementary angles is 180°, so a = 180 – 64 = 116.

Choice A is incorrect and likely results from thinking the angle with measure a° is the complement of the angle with measure 64°. Choice B is incorrect and likely results from thinking the angle with measure a° is congruent to the angle with measure 64°. Choice D is incorrect and likely results from a conceptual or computational error.

▼ 1.15 — ?

Choice C is correct. There is a proportional relationship between the height of an object and the length of its shadow. Let c represent the height, in feet, of the cherry tree. The given relationship can be expressed by the proportion $\frac{555}{300}=\frac{c}{16}$. Multiplying both sides of this equation by 16 yields $c=29.6$. This height is closest to the value given in choice C, 30.

Choices A, B, and D are incorrect and may result from calculation errors.

▼ 1.16 — A

Choice A is correct. Given that lines m and n are parallel, the angle marked 130° must be supplementary to the leftmost angle marked a° because they are same-side interior angles. Therefore, 130° + a° = 180°, which yields a = 50°. Lines ℓ and m intersect at a right angle, so lines j, ℓ, and m form a right triangle where the two acute angles are a° and b°. The acute angles of a right triangle are complementary, so a° + b° = 90°, which yields 50° + b° = 90°, and b = 40.

Choice B is incorrect. This is the value of a, not b. Choice C is incorrect and may be the result of dividing 130° by 2. Choice D is incorrect and may be the result of multiplying b by 2.

▼ 1.17 — D

Choice D is correct. It's given that lines *m* and *n* are parallel. Since line *t* intersects both lines *m* and *n*, it's a transversal. The angles in the figure marked as $170°$ and $w°$ are on the same side of the transversal, where one is an interior angle with line *m* as a side, and the other is an exterior angle with line *n* as a side. Thus, the marked angles are corresponding angles. When two parallel lines are intersected by a transversal, corresponding angles are congruent and, therefore, have equal measure. It follows that $w° = 170°$. Therefore, the value of *w* is 170.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.18 — D

Choice D is correct. Since $\angle BCD$ and $\angle BCA$ form a linear pair of angles, their measures sum to 180°. It's given that the measure of $\angle BCD$ is 110°. Therefore, $110° + \angle BCA = 180°$. Subtracting 110° from both sides of this equation gives the measure of $\angle BCA$ as 70°. It's also given that the measure of $\angle BAC$ is equal to the measure of $\angle BCA$. Thus, the measure of $\angle BAC$ is also 70°. The measures of the interior angles of a triangle sum to 180°. Thus, $70° + 70° + x° = 180°$. Combining like terms on the left-hand side of this equation yields $140° + x° = 180°$. Subtracting 140° from both sides of this equation yields $x° = 40°$, or $x = 40$.

Choice A is incorrect. This is the value of the measure of $\angle BCD$. Choice B is incorrect. This is the value of the measure of each of the other two interior angles, $\angle BCA$ and $\angle BAC$. Choice C is incorrect and may result from an error made when identifying the relationship between the exterior angle of a triangle and the interior angles of the triangle.

▼ 1.19 — A

Choice A is correct. It's given that *BC* is parallel to *AD*. Since line *t* intersects $\overline{BC}$ and $\overline{AD}$, it's a transversal. The angles in the figure marked as *x*° and *y*° are on opposite sides of the transversal and lie between the parallel segments, so they are alternate interior angles. When two parallel segments are intersected by a transversal, alternate interior angles have equal measure. It's also given that $x = 22$. Therefore, the value of *y* is 22.

Choice B is incorrect. This is the value of $2x$, not the value of *y*.

Choice C is incorrect. This is the complement of 22, not the value of *y*.

Choice D is incorrect. This is the supplement of 22 , not the value of *y*.

▼ 1.20 — B

Choice B is correct. It's given that triangles *EFG* and *JKL* are congruent such that angle *E* corresponds to angle *J*. Corresponding angles of congruent triangles are congruent, so angle *E* and angle *J* must be congruent. Therefore, if the measure of angle *E* is $45°$, then the measure of angle *J* is also $45°$.

Choice A is incorrect. This is the measure of angle *K*, not angle *J*.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.21 — B

Choice B is correct. The sum of the interior angles of a triangle is 180°. The measures of the two interior angles of the given triangle are shown. Therefore, the measure of the third interior angle is 180° – 70° – 50° = 60°. The angles of measures x° and 60° are supplementary, so their sum is 180°. Therefore, x = 180 – 60 = 120.

Choice A is incorrect and may be the result of misinterpreting x° as supplementary to 70°. Choice C is incorrect and may be the result of misinterpreting x° as supplementary to 50°.

Choice D is incorrect and may be the result of a calculation error.

▼ 1.22 — 175

The correct answer is 175. It's given that line *g* is parallel to line *h*. Since line *l* intersects both lines, it's a transversal. The angles in the figure marked as $175°$ and $x°$ are on the same side of the transversal, where one is an interior angle with line *h* as a side, and the other is an exterior angle with line *g* as a side, respectively. Thus, the marked angles are corresponding angles. When two parallel lines are intersected by a transversal, corresponding angles have equal measure. Therefore, the value of *x* is 175.

▼ 1.23 — D

Choice D is correct. The sum of the angle measures of a triangle is $180°$. It's given that the measure of angle *Q* is $132°$, the measure of angle *P* is $24°$, and the measure of angle *R* is $x°$. It follows that $x + 132 + 24 = 180$, or $x + 152 = 180$. Subtracting 152 from both sides of this equation yields $x = 24$.

Alternate approach: It's given that in the triangle shown, $PQ = QR$. Thus, triangle *PQR* is an isosceles triangle. Since base angles of an isosceles triangle are congruent, the measure of angle *P* is equal to the measure of angle *R*. Therefore, the measure of angle *R* is $24°$, and the value of *x* is 24.

Choice A is incorrect. This is the sum of the measures, in degrees, of angles *Q* and *P*, not the measure, in degrees, of angle *R*.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the sum of the measures, in degrees, of angles *P* and *R*, not the measure, in degrees, of angle *R*.

▼ 1.24 — C

Choice C is correct. Vertical angles, or angles that are opposite each other when two lines intersect, are congruent. It's given that line *k* intersects line *n*. Based on the figure, the angle with measure $x°$ and the angle with measure $145°$ are vertical angles. Therefore, the value of *x* is equal to 145.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.25 — B

Choice B is correct. Let the measure of the third angle in the smaller triangle be $a°$. Since lines ℓ and m are parallel and cut by transversals, it follows that the corresponding angles formed are congruent. So $a° = y° = 20°$. The sum of the measures of the interior angles of a triangle is $180°$, which for the interior angles in the smaller triangle yields $a + x + z = 180$. Given that $z = 60$ and $a = 20$, it follows that $20 + x + 60 = 180$. Solving for x gives $x = 180 - 60 - 20$, or $x = 100$.

Choice A is incorrect and may result from incorrectly assuming that angles $x + z = 180$. Choice C is incorrect and may result from incorrectly assuming that the smaller triangle is a right triangle, with x as the right angle. Choice D is incorrect and may result from a misunderstanding of the exterior angle theorem and incorrectly assuming that $x = y + z$.

▼ 1.26 — A

Choice A is correct. It's given that angle 1 and angle 2 are vertical angles, and the measure of angle 1 is $72°$. Vertical angles have equal measures. Therefore, the measure of angle 2 is $72°$.

Choice B is incorrect. This is the measure of an angle that is supplementary, not congruent, to angle 1.

Choice C is incorrect. This is the sum of the measures of angle 1 and angle 2.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.27 — D

Choice D is correct. The sum of consecutive interior angles between two parallel lines and on the same side of the transversal is 180 degrees. Since it's given that line *m* is parallel to line *n*, it follows that $x + 26 = 180$. Subtracting 26 from both sides of this equation yields 154. Therefore, the value of x is 154.

Choice A is incorrect. This is half of the given angle measure.

Choice B is incorrect. This is the value of the given angle measure.

Choice C is incorrect. This is twice the value of the given angle measure.

▼ 1.28 — A

Choice A is correct. The sum of the measures of the angles of a triangle is $180°$. Therefore, the sum of the measures of $\angle R$, $\angle S$, and $\angle T$ is $180°$. It's given that the measure of $\angle R$ is $63°$. It follows that the sum of the measures of $\angle S$ and $\angle T$ is $180 - 63°$, or $117°$. Therefore, the measure of $\angle S$, in degrees, must be less than 117. Of the given choices, only 116 is less than 117. Thus, the measure, in degrees, of $\angle S$ could be 116.

Choice B is incorrect. If the measure of $\angle S$ is $118°$, then the sum of the measures of the angles of the triangle is greater than, not equal to, $180°$.

Choice C is incorrect. If the measure of $\angle S$ is $126°$, then the sum of the measures of the angles of the triangle is greater than, not equal to, $180°$.

Choice D is incorrect. This is the sum of the measures of the angles of a triangle, in degrees.

▼ 1.29 — B

Choice B is correct. Since $AB = AC$, the measures of their corresponding angles, $\angle ABC$ and $\angle ACB$, are equal. Since $\angle ABC$ has a measure of $67°$, the measure of $\angle ACB$ is also $67°$. Since the sum of the measures of the interior angles in a triangle is $180°$, it follows that $67 + 67 + x = 180$, or $134 + x = 180$. Subtracting by 134 on both sides of this equation yields $x = 46$.

Choices A, C, and D are incorrect and may result from calculation errors.

▼ 1.30 — D

Choice D is correct. It's given that lines m and n are parallel. Since line t intersects both lines m and n, it's a transversal. The angles in the figure marked as $134°$ and $w°$ are on the same side of the transversal, where one is an interior angle with line m as a side, and the other is an exterior angle with line n as a side. Thus, the marked angles are corresponding angles. When two parallel lines are intersected by a transversal, corresponding angles are congruent and, therefore, have equal measure. It follows that $w° = 134°$. Therefore, the value of w is 134.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.31 — A

Choice A is correct. The triangle angle sum theorem states that the sum of the measures of the interior angles of a triangle is $180°$. It's given that in $\triangle XYZ$, the measure of $\angle X$ is $24°$ and the measure of $\angle Y$ is $98°$. It follows that the measure of $\angle Z$ is $180 - 24 - 98°$, or $58°$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the sum of the measures of $\angle X$ and $\angle Y$, not the measure of $\angle Z$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.32 — 70

The correct answer is 70. Based on the figure, the angle with measure $110°$ and the angle vertical to the angle with measure $x°$ are same side interior angles. Since vertical angles are congruent, the angle vertical to the angle with measure $x°$ also has measure $x°$. It's given that lines s and t are parallel. Therefore, same side interior angles between lines s and t are supplementary. It follows that $x + 110 = 180$. Subtracting 110 from both sides of this equation yields $x = 70$.

▼ 1.33 — B

Choice B is correct. In similar triangles, corresponding angles are congruent. It's given that right triangles PQR and STU are similar, where angle P corresponds to angle S. It follows that angle P is congruent to angle S. In the triangles shown, angle R and angle U are both marked as right angles, so angle R and angle U are corresponding angles. It follows that angle Q and angle T are corresponding angles, and thus, angle Q is congruent to angle T. It's given that the measure of angle Q is $18°$, so the measure of angle T is also $18°$. Angle U is a right angle, so the measure of angle U is $90°$. The sum of the measures of the interior angles of a triangle is $180°$. Thus, the sum of the measures of the interior angles of triangle STU is 180 degrees. Let s represent the measure, in degrees, of angle S. It follows that $s + 18 + 90 = 180$, or $s + 108 = 180$. Subtracting 108 from both sides of this equation yields $s = 72$. Therefore, if the measure of angle Q is 18 degrees, then the measure of angle S is 72 degrees.

Choice A is incorrect. This is the measure of angle T.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the sum of the measures of angle S and angle U.

▼ 1.34 — B

Choice B is correct. In the figure shown, transversal t intersects lines r and s. Two lines intersected by a transversal are parallel if and only if a pair of alternate interior angles have equal measure. The angle marked $24°$ and the angle marked $x°$ form a pair of alternate interior angles. Therefore, $x = 24$ is sufficient to prove that lines r and s are parallel.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 1.35 — B

Choice B is correct. It's given that two lines intersect at a point. The angles with measures of $r°$ and $s°$ are vertical angles, so it follows that they are equal in measure. Therefore, $s = r$. If $r = 53$, it follows that $s = 53$.

Choice A is incorrect. This is the measure, in degrees, of the complement of an angle with a measure of $53°$, not the value of s.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.36 — C

Choice C is correct. It's given that $x = 100$; therefore, substituting 100 for x in triangle ABC gives two known angle measures for this triangle. The sum of the measures of the interior angles of any triangle equals 180°. Subtracting the two known angle measures of triangle ABC from 180° gives the third angle measure: $180° - 100° - 20° = 60°$. This is the measure of angle BCA. Since vertical angles are congruent, the measure of angle DCE is also 60°. Subtracting the two known angle measures of triangle CDE from 180° gives the third angle measure: $180° - 60° - 40° = 80°$. Therefore, the value of y is 80.

Choice A is incorrect and may result from a calculation error. Choice B is incorrect and may result from classifying angle CDE as a right angle. Choice D is incorrect and may result from finding the measure of angle BCA or DCE instead of the measure of angle CDE.

▼ 1.37 — C

Choice C is correct. It's given that triangle XYZ is obtained by a dilation of triangle ABC. It follows that triangle ABC is similar to triangle XYZ, where A corresponds to X. Since corresponding angles in similar triangles have the same measure and the measure of angle A is 60 degrees, it follows that the measure of angle X is also 60 degrees.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 1.38 — A

Choice A is correct. The sum of the measures of the three angles in a triangle is $180°$. It's given that in $\triangle JKL$, the measure of $\angle L$ is $118°$. Let x represent the measure of $\angle J$. It's also given that the measures of $\angle J$ and $\angle K$ are equal, so x also represents the measure of $\angle K$. It follows that $x + x + 118 = 180$, or $2x + 118 = 180$. Subtracting 118 from both sides of this equation yields $2x = 62$. Dividing both sides of this equation by 2 yields $x = 31$. Therefore, the measure of $\angle J$ is $31°$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the sum of the measures of $\angle J$ and $\angle K$, not the measure of $\angle J$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.39 — C

Choice C is correct. The triangle angle sum theorem states that the sum of the measures of the interior angles of a triangle is $180°$. It's given that in $\triangle XYZ$, the measure of $\angle X$ is $23°$ and the measure of $\angle Y$ is $66°$. It follows that the measure of $\angle Z$ is $180 - 23 - 66°$, or $91°$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the sum of the measures of $\angle X$ and $\angle Y$, not the measure of $\angle Z$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.40 — D

Choice D is correct. The sum of the angle measures of a triangle is $180°$. Adding the measures of angles B and C gives $52 + 17 = 69°$. Therefore, the measure of angle A is $180 - 69 = 111°$.

Choice A is incorrect and may result from subtracting the sum of the measures of angles B and C from $90°$, instead of from $180°$.

Choice B is incorrect and may result from subtracting the measure of angle C from the measure of angle B.

Choice C is incorrect and may result from adding the measures of angles B and C but not subtracting the result from $180°$.

▼ 1.41 — ?

The correct answer is 20. By the equality given, the three pairs of corresponding sides of the two triangles are in the same proportion. By the side-side-side (SSS) similarity theorem, triangle ABC is similar to triangle DEF. In similar triangles, the measures of corresponding angles are congruent. Since angle BAC corresponds to angle EDF, these two angles are congruent and their measures are equal. It's given that the measure of angle BAC is 20°, so the measure of angle EDF is also 20°.

▼ 1.42 — C

Choice C is correct. Vertical angles, which are angles that are opposite each other when two lines intersect, are congruent. The figure shows that lines t and m intersect. It follows that the angle with measure $x°$ and the angle with measure $y°$ are vertical angles, so $x = y$. It's given that $x = 6k + 13$ and $y = 8k - 29$. Substituting $6k + 13$ for x and $8k - 29$ for y in the equation $x = y$ yields $6k + 13 = 8k - 29$. Subtracting $6k$ from both sides of this equation yields $13 = 2k - 29$. Adding 29 to both sides of this equation yields $42 = 2k$, or $2k = 42$. Dividing both sides of this equation by 2 yields $k = 21$. It's given that lines m and n are parallel, and the figure shows that lines m and n are intersected by a transversal, line t. If two parallel lines are intersected by a transversal, then the same-side interior angles are supplementary. It

follows that the same-side interior angles with measures $y°$ and $z°$ are supplementary, so $y + z = 180$. Substituting $8k - 29$ for y in this equation yields $8k - 29 + z = 180$. Substituting 21 for k in this equation yields $821 - 29 + z = 180$, or $139 + z = 180$. Subtracting 139 from both sides of this equation yields $z = 41$. Therefore, the value of z is 41.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of k, not z.

Choice D is incorrect. This is the value of x or y, not z.

▼ 1.43 — C

Choice C is correct. It's given that triangle *CAE* is similar to triangle *CBD*. Corresponding angles in similar triangles have equal measure. Angle *BCD* and angle *ACE* represent the same angle. It follows that angle *BCD* and angle *ACE* have equal measure and are corresponding angles. It's given in the figure that angle *BDC* and angle *AEC* are right angles and therefore have equal measure. It follows that angle *BDC* and angle *AEC* are corresponding angles. Therefore, angle *CBD* and angle *CAE* are corresponding angles and have equal measure. It's given that the measure of angle *CBD* is $57°$, so the measure of angle *CAE* is $57°$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 1.44 — A

Choice A is correct. It’s given that *AE* intersects *DB* at point *C* and that *AB* is parallel to *DE*, creating triangles *ABC* and *CDE*. Since *AB* is parallel to *DE*, angles *BAC* and *DEC* are opposite interior angles and are congruent. It follows that the measure of angle *DEC* is $y°$. This means the three interior angles of triangle *CDE* have measures of $x°$, $z°$, and $y°$, respectively. It’s given that $x = 63$ and $y = 2x - 47$. Substituting 63 for x in the equation $y = 2x - 47$ yields $y = 2(63) - 47$, or $y = 79$. Therefore, the three interior angles of triangle *CDE* have measures $63°$, $z°$, and $79°$, respectively. The sum of the interior angles of a triangle is $180°$. Therefore, $63 + z + 79 = 180$, or $z + 142 = 180$. Subtracting 142 from both sides of this equation yields $z = 38$. Therefore, the value of z is 38.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of x, not z.

Choice D is incorrect. This is the value of y, not z.

▼ 1.45 — B

Choice B is correct. It's given that triangle LMN is similar to triangle PQR. Corresponding angles of similar triangles are congruent. Since angle M and angle Q correspond to each other, they must be congruent. Therefore, if the measure of angle M is $53°$, then the measure of angle Q is also $53°$.

Choice A is incorrect and may result from concluding that angle M and angle Q are complementary rather than congruent.

Choice C is incorrect and may result from concluding that angle M and angle Q are supplementary rather than congruent.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.46 — B

Choice B is correct. It's given that angle ACE measures $62°$. Since segments AE and BD are parallel, angles BDC and CEA are congruent. Therefore, angle CEA measures $58°$. The sum of the measures of angles ACE, CEA, and CAE is $180°$ since the sum of the interior angles of triangle ACE is equal to $180°$. Let the measure of angle CAE be $x°$. Therefore, $62+58+x=180$, which simplifies to $x=60$. Thus, the measure of angle CAE is $60°$.

Choice A is incorrect. This is the measure of angle AEC, not that of angle CAE. Choice C is incorrect. This is the measure of angle ACE, not that of CAE. Choice D is incorrect. This is the sum of the measures of angles ACE and CEA.

▼ 1.47 — 21

The correct answer is 21. It's given that triangles ABC and DEF are congruent with angle A corresponding to angle D. Corresponding angles of congruent triangles are congruent and, therefore, have equal measure. It's given that the measure of angle A is $69°$. It follows that the measure of angle D is also $69°$. It's given that angle E is a right angle. Therefore, the measure of angle E is $90°$. Let x represent the measure, in degrees, of angle F. Since the measures of the angles in a triangle sum to $180°$, it follows that $69 + 90 + x = 180$, or

$159 + x = 180$. Subtracting 159 from both sides of this equation yields $x = 21$. Therefore, the measure, in degrees, of angle F is 21.

▼ 1.48 — A

Choice A is correct. In the figure shown, line t, which is a transversal, intersects lines j and k. If two lines are intersected by a transversal and the measures of a pair of same-side interior angles sum to $180°$, then the lines are parallel. In the figure, the angle measuring $134°$ and the angle measuring $x°$ are a pair of same-side interior angles. If $x = 46$, $(134 + x)° = (134 + 46)°$, or $(134 + x)° = 180°$. Therefore, $x = 46$ is sufficient to prove that lines j and k are parallel.

Choice B is incorrect. The angle measuring $y°$ and the angle measuring $134°$ are alternate interior angles. If $y = 46$ rather than $y = 134$, then lines j and k would not be parallel.

Choice C is incorrect. The angle measuring $w°$ and the angle measuring $134°$ are vertical angles. Thus, $w = 134$ whether lines j and k are parallel or not.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.49 — B

Choice B is correct. In the figure shown, since $\bar{QS}$ is a line segment, the sum of the measures of $\angle PRS$ and $\angle PRQ$ is $180°$. It's given that the measure of $\angle PRS$ is $163°$. Thus, the measure of $\angle PRQ$ is $(180 - 163)°$, or $17°$. The sum of the measures of the interior angles of a triangle is $180°$. It's given that the measure of $\angle PQR$ is $132°$. Therefore, the measure of $\angle QPR$ is $(180 - 17 - 132)°$, or $31°$.

Choice A is incorrect. This is the measure of the supplement of $\angle PQR$, not the measure of $\angle QPR$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the measure of $\angle PRQ$, not the measure of $\angle QPR$.

▼ 1.50 — A

Choice A is correct. The figure shows that angle P in $\triangle QPR$ and angle T in $\triangle STR$ are right angles. It follows that angle P is congruent to angle T. The figure also shows that the measures of angle QRP and angle SRT are both $x°$. Therefore, angle QRP is congruent to angle SRT. It's given that $\triangle QPR$ is similar to $\triangle STR$. Since angle P is congruent to angle

T, and angle QRP is congruent to angle SRT, it follows that $\bar{QR}$ corresponds to $\bar{SR}$, and $\bar{QP}$ corresponds to $\bar{ST}$. Since corresponding sides of similar triangles are proportional, it follows that $\frac{SR}{QR} = \frac{ST}{QP}$. It's also given that the lengths of $\bar{ST}$, $\bar{QP}$, and $\bar{QR}$ are 14, 15, and 25, respectively. Substituting 14 for ST, 15 for QP, and 25 for QR in the equation $\frac{SR}{QR} = \frac{ST}{QP}$ yields $\frac{SR}{25} = \frac{14}{15}$. Multiplying each side of this equation by 25 yields $SR = \frac{14}{15}25$, or $SR = \frac{350}{15}$. Thus, the length of $\bar{SR}$ is $\frac{350}{15}$.

Choice B is incorrect. This is the result of solving the equation $\frac{SR}{25} = \frac{14}{20}$, not $\frac{SR}{25} = \frac{14}{15}$.

Choice C is incorrect. This is the result of solving the equation $\frac{SR}{14} = \frac{15}{20}$, not $\frac{SR}{25} = \frac{14}{15}$.

Choice D is incorrect. This is the result of solving the equation $\frac{SR}{14} = \frac{15}{25}$, not $\frac{SR}{25} = \frac{14}{15}$.

▼ 1.51 — D

Choice D is correct. It's given that triangle ABC is similar to triangle XYZ, where A, B, and C correspond to X, Y, and Z, respectively. It follows that side AB corresponds to side XY and side BC corresponds to side YZ. Since the lengths of corresponding sides in similar triangles are proportional, it follows that $\frac{XY}{AB} = \frac{YZ}{BC}$. Substituting 170 for AB, 60 for YZ, and 850 for BC in this equation yields $\frac{XY}{170} = \frac{60}{850}$. Multiplying each side of this equation by 170 yields $XY = 12$. Therefore, the length of $\bar{XY}$ is 12.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the length of $\bar{YZ}$, not $\bar{XY}$.